4697566D My Rental Price SageMaker Pipeline V2

Singapore non-landed rental price estimation using XGBoost

The pipeline covers data cleaning, processing, training, evaluation and conditional registration flow, but incorporates the successful MLflow deployment pattern: a SageMaker `Model` is created with the trained artifact plus custom `inference.py`, and the model is registered. This lets the registered model deploy with the custom inference specification.

In [1]:
```
# After running, restart the kernel before continuing if packages were upgraded.
%pip install --upgrade "sagemaker>=2,<3" boto3 botocore mlflow sagemaker-mlflow
```

```
Collecting sagemaker<3,>=2
  Using cached sagemaker-2.257.6-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: boto3 in /opt/conda/lib/python3.12/site-packages (1.43.46)
Collecting boto3
  Downloading boto3-1.43.78-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: botocore in /opt/conda/lib/python3.12/site-packages (1.43.46)
Collecting botocore
  Downloading botocore-1.43.78-py3-none-any.whl.metadata (5.6 kB)
Requirement already satisfied: mlflow in /opt/conda/lib/python3.12/site-packages (3.13.0)
Collecting mlflow
  Using cached mlflow-3.15.1-py3-none-any.whl.metadata (49 kB)
Requirement already satisfied: sagemaker-mlflow in /opt/conda/lib/python3.12/site-packages (0.5.0)
Collecting attrs<26,>=24 (from sagemaker<3,>=2)
  Using cached attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: cloudpickle>=2.2.1 in /opt/conda/lib/python3.12/site-packages (from sagema
ker<3,>=2) (3.1.2)
Requirement already satisfied: docker in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(7.2.0)
Requirement already satisfied: fastapi in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(0.139.2)
Requirement already satisfied: google-pasta in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,
>=2) (0.2.0)
Requirement already satisfied: graphene<4,>=3 in /opt/conda/lib/python3.12/site-packages (from sagemaker<
3,>=2) (3.4.3)
Requirement already satisfied: importlib-metadata<7.0,>=1.4.0 in /opt/conda/lib/python3.12/site-packages
(from sagemaker<3,>=2) (6.10.0)
Requirement already satisfied: jsonschema in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=
2) (4.23.0)
Requirement already satisfied: numpy<3.0,>=1.26.4 in /opt/conda/lib/python3.12/site-packages (from sagema
ker<3,>=2) (1.26.4)
Requirement already satisfied: omegaconf<3,>=2.2 in /opt/conda/lib/python3.12/site-packages (from sagemak
er<3,>=2) (2.3.0)
Collecting packaging<25,>=23.0 (from sagemaker<3,>=2)
  Using cached packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: pandas>=2.3.0 in /opt/conda/lib/python3.12/site-packages (from sagemaker<
3,>=2) (2.3.3)
Collecting pathos (from sagemaker<3,>=2)
  Using cached pathos-0.3.5-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: platformdirs in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,
>=2) (4.11.0)
Requirement already satisfied: protobuf<7.0,>=3.12 in /opt/conda/lib/python3.12/site-packages (from sagem
aker<3,>=2) (6.31.1)
Requirement already satisfied: psutil in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(5.9.8)
Requirement already satisfied: pytz in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2) (20
24.2)
Requirement already satisfied: pyyaml>=6.0.1 in /opt/conda/lib/python3.12/site-packages (from sagemaker<
3,>=2) (6.0.3)
Requirement already satisfied: requests in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(2.34.2)
Collecting sagemaker-core<2.0.0,>=1.0.71 (from sagemaker<3,>=2)
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  Using cached sagemaker_core-1.0.78-py3-none-any.whl.metadata (4.9 kB)
Requirement already satisfied: schema in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(0.7.8)
Requirement already satisfied: smdebug-rulesconfig==1.0.1 in /opt/conda/lib/python3.12/site-packages (fro
m sagemaker<3,>=2) (1.0.1)
Requirement already satisfied: tblib<4,>=1.7.0 in /opt/conda/lib/python3.12/site-packages (from sagemaker
<3,>=2) (3.2.2)
Requirement already satisfied: tqdm in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2) (4.
69.0)
Requirement already satisfied: urllib3<3.0.0,>=1.26.8 in /opt/conda/lib/python3.12/site-packages (from sa
gemaker<3,>=2) (1.26.20)
Requirement already satisfied: uvicorn in /opt/conda/lib/python3.12/site-packages (from sagemaker<3,>=2)
(0.51.0)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /opt/conda/lib/python3.12/site-packages (from bo
to3) (1.1.0)
Requirement already satisfied: s3transfer<0.20.0,>=0.19.0 in /opt/conda/lib/python3.12/site-packages (fro
m boto3) (0.19.1)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /opt/conda/lib/python3.12/site-packages (fr
om botocore) (2.9.0.post0)
Requirement already satisfied: graphql-core<3.3,>=3.1 in /opt/conda/lib/python3.12/site-packages (from gr
aphene<4,>=3->sagemaker<3,>=2) (3.2.11)
Requirement already satisfied: graphql-relay<3.3,>=3.1 in /opt/conda/lib/python3.12/site-packages (from g
raphene<4,>=3->sagemaker<3,>=2) (3.2.0)
Requirement already satisfied: typing-extensions<5,>=4.7.1 in /opt/conda/lib/python3.12/site-packages (fr
om graphene<4,>=3->sagemaker<3,>=2) (4.16.0)
Requirement already satisfied: zipp>=0.5 in /opt/conda/lib/python3.12/site-packages (from importlib-metad
ata<7.0,>=1.4.0->sagemaker<3,>=2) (4.1.0)
Requirement already satisfied: antlr4-python3-runtime==4.9.* in /opt/conda/lib/python3.12/site-packages
(from omegaconf<3,>=2.2->sagemaker<3,>=2) (4.9.3)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.12/site-packages (from python-dateutil<
3.0.0,>=2.1->botocore) (1.17.0)
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in /opt/conda/lib/python3.12/site-packages (from sa
gemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (2.13.4)
Requirement already satisfied: rich<15.0.0,>=13.0.0 in /opt/conda/lib/python3.12/site-packages (from sage
maker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (14.3.4)
Requirement already satisfied: mock<5.0,>4.0 in /opt/conda/lib/python3.12/site-packages (from sagemaker-c
ore<2.0.0,>=1.0.71->sagemaker<3,>=2) (4.0.3)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /opt/conda/lib/python3.12/site-pac
kages (from jsonschema->sagemaker<3,>=2) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /opt/conda/lib/python3.12/site-packages (from jsons
chema->sagemaker<3,>=2) (0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in /opt/conda/lib/python3.12/site-packages (from jsonschema
->sagemaker<3,>=2) (0.27.1)
Requirement already satisfied: annotated-types>=0.6.0 in /opt/conda/lib/python3.12/site-packages (from py
dantic<3.0.0,>=2.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in /opt/conda/lib/python3.12/site-packages (from pyd
antic<3.0.0,>=2.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /opt/conda/lib/python3.12/site-packages (from
pydantic<3.0.0,>=2.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (0.4.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/conda/lib/python3.12/site-packages (from ric
h<15.0.0,>=13.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /opt/conda/lib/python3.12/site-packages (from r
ich<15.0.0,>=13.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (2.20.0)
Collecting mlflow-skinny==3.15.1 (from mlflow)
  Using cached mlflow_skinny-3.15.1-py3-none-any.whl.metadata (50 kB)
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Collecting mlflow-tracing==3.15.1 (from mlflow)
  Using cached mlflow_tracing-3.15.1-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: Flask-CORS<7 in /opt/conda/lib/python3.12/site-packages (from mlflow) (6.
0.5)
Requirement already satisfied: Flask<4 in /opt/conda/lib/python3.12/site-packages (from mlflow) (3.1.3)
Requirement already satisfied: aiohttp<4,>=3.7.0 in /opt/conda/lib/python3.12/site-packages (from mlflow)
(3.14.2)
Requirement already satisfied: alembic!=1.10.0,<2 in /opt/conda/lib/python3.12/site-packages (from mlflo
w) (1.18.5)
Requirement already satisfied: cryptography<50,>=43.0.0 in /opt/conda/lib/python3.12/site-packages (from
mlflow) (46.0.7)
Requirement already satisfied: gunicorn<27 in /opt/conda/lib/python3.12/site-packages (from mlflow) (23.
0.0)
Requirement already satisfied: huey<4,>=2.5.4 in /opt/conda/lib/python3.12/site-packages (from mlflow)
(2.6.0)
Requirement already satisfied: matplotlib<4 in /opt/conda/lib/python3.12/site-packages (from mlflow) (3.1
0.9)
Requirement already satisfied: pyarrow<26,>=4.0.0 in /opt/conda/lib/python3.12/site-packages (from mlflo
w) (21.0.0)
Requirement already satisfied: scikit-learn<2 in /opt/conda/lib/python3.12/site-packages (from mlflow)
(1.7.2)
Requirement already satisfied: scipy<2 in /opt/conda/lib/python3.12/site-packages (from mlflow) (1.16.3)
Requirement already satisfied: skops<1 in /opt/conda/lib/python3.12/site-packages (from mlflow) (0.14.0)
Requirement already satisfied: sqlalchemy<3,>=1.4.0 in /opt/conda/lib/python3.12/site-packages (from mlfl
ow) (2.0.51)
Requirement already satisfied: cachetools<8,>=5.0.0 in /opt/conda/lib/python3.12/site-packages (from mlfl
ow-skinny==3.15.1->mlflow) (6.2.6)
Requirement already satisfied: click<9,>=7.0 in /opt/conda/lib/python3.12/site-packages (from mlflow-skin
ny==3.15.1->mlflow) (8.2.1)
Requirement already satisfied: databricks-sdk<1,>=0.20.0 in /opt/conda/lib/python3.12/site-packages (from
mlflow-skinny==3.15.1->mlflow) (0.122.0)
Requirement already satisfied: gitpython<4,>=3.1.9 in /opt/conda/lib/python3.12/site-packages (from mlflo
w-skinny==3.15.1->mlflow) (3.1.53)
Requirement already satisfied: opentelemetry-api<3,>=1.9.0 in /opt/conda/lib/python3.12/site-packages (fr
om mlflow-skinny==3.15.1->mlflow) (1.43.0)
Requirement already satisfied: opentelemetry-proto<3,>=1.9.0 in /opt/conda/lib/python3.12/site-packages
(from mlflow-skinny==3.15.1->mlflow) (1.44.0)
Requirement already satisfied: opentelemetry-sdk<3,>=1.9.0 in /opt/conda/lib/python3.12/site-packages (fr
om mlflow-skinny==3.15.1->mlflow) (1.43.0)
Requirement already satisfied: python-dotenv<2,>=0.19.0 in /opt/conda/lib/python3.12/site-packages (from
mlflow-skinny==3.15.1->mlflow) (1.2.2)
Requirement already satisfied: sqlparse<1,>=0.4.0 in /opt/conda/lib/python3.12/site-packages (from mlflow
-skinny==3.15.1->mlflow) (0.5.0)
Requirement already satisfied: starlette<2 in /opt/conda/lib/python3.12/site-packages (from mlflow-skinny
==3.15.1->mlflow) (0.52.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /opt/conda/lib/python3.12/site-packages (from a
iohttp<4,>=3.7.0->mlflow) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in /opt/conda/lib/python3.12/site-packages (from aiohttp<
4,>=3.7.0->mlflow) (1.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in /opt/conda/lib/python3.12/site-packages (from aiohttp
<4,>=3.7.0->mlflow) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /opt/conda/lib/python3.12/site-packages (from aioht
tp<4,>=3.7.0->mlflow) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in /opt/conda/lib/python3.12/site-packages (from aiohttp<
4,>=3.7.0->mlflow) (0.5.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /opt/conda/lib/python3.12/site-packages (from aiohttp
<4,>=3.7.0->mlflow) (1.24.5)
Requirement already satisfied: Mako in /opt/conda/lib/python3.12/site-packages (from alembic!=1.10.0,<2->
mlflow) (1.3.12)
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Requirement already satisfied: cffi>=1.14 in /opt/conda/lib/python3.12/site-packages (from cryptography<5
0,>=43.0.0->mlflow) (1.17.1)
Requirement already satisfied: google-auth~=2.0 in /opt/conda/lib/python3.12/site-packages (from databric
ks-sdk<1,>=0.20.0->mlflow-skinny==3.15.1->mlflow) (2.56.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /opt/conda/lib/python3.12/site-packages (from fast
api->sagemaker<3,>=2) (0.0.4)
Requirement already satisfied: blinker>=1.9.0 in /opt/conda/lib/python3.12/site-packages (from Flask<4->m
lflow) (1.9.0)
Requirement already satisfied: itsdangerous>=2.2.0 in /opt/conda/lib/python3.12/site-packages (from Flask
<4->mlflow) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in /opt/conda/lib/python3.12/site-packages (from Flask<4->ml
flow) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in /opt/conda/lib/python3.12/site-packages (from Flask<4
->mlflow) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in /opt/conda/lib/python3.12/site-packages (from Flask<4->
mlflow) (3.1.8)
Requirement already satisfied: gitdb<5,>=4.0.1 in /opt/conda/lib/python3.12/site-packages (from gitpython
<4,>=3.1.9->mlflow-skinny==3.15.1->mlflow) (4.0.12)
Requirement already satisfied: smmap<6,>=3.0.1 in /opt/conda/lib/python3.12/site-packages (from gitdb<5,>
=4.0.1->gitpython<4,>=3.1.9->mlflow-skinny==3.15.1->mlflow) (5.0.3)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /opt/conda/lib/python3.12/site-packages (from goo
gle-auth~=2.0->databricks-sdk<1,>=0.20.0->mlflow-skinny==3.15.1->mlflow) (0.4.2)
Requirement already satisfied: contourpy>=1.0.1 in /opt/conda/lib/python3.12/site-packages (from matplotl
ib<4->mlflow) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /opt/conda/lib/python3.12/site-packages (from matplotlib<4
->mlflow) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /opt/conda/lib/python3.12/site-packages (from matplot
lib<4->mlflow) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /opt/conda/lib/python3.12/site-packages (from matplot
lib<4->mlflow) (1.5.0)
Requirement already satisfied: pillow>=8 in /opt/conda/lib/python3.12/site-packages (from matplotlib<4->m
lflow) (11.3.0)
Requirement already satisfied: pyparsing>=3 in /opt/conda/lib/python3.12/site-packages (from matplotlib<4
->mlflow) (3.3.2)
Requirement already satisfied: opentelemetry-semantic-conventions==0.64b0 in /opt/conda/lib/python3.12/si
te-packages (from opentelemetry-sdk<3,>=1.9.0->mlflow-skinny==3.15.1->mlflow) (0.64b0)
Requirement already satisfied: tzdata>=2022.7 in /opt/conda/lib/python3.12/site-packages (from pandas>=2.
3.0->sagemaker<3,>=2) (2026.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/conda/lib/python3.12/site-packages (from
requests->sagemaker<3,>=2) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.12/site-packages (from requests->sa
gemaker<3,>=2) (3.18)
Requirement already satisfied: certifi>=2023.5.7 in /opt/conda/lib/python3.12/site-packages (from request
s->sagemaker<3,>=2) (2026.6.17)
Requirement already satisfied: joblib>=1.2.0 in /opt/conda/lib/python3.12/site-packages (from scikit-lear
n<2->mlflow) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /opt/conda/lib/python3.12/site-packages (from scik
it-learn<2->mlflow) (3.6.0)
Collecting prettytable>=3.9 (from skops<1->mlflow)
  Using cached prettytable-3.18.0-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: greenlet>=1 in /opt/conda/lib/python3.12/site-packages (from sqlalchemy<3,
>=1.4.0->mlflow) (3.5.3)
Requirement already satisfied: anyio<5,>=3.6.2 in /opt/conda/lib/python3.12/site-packages (from starlette
<2->mlflow-skinny==3.15.1->mlflow) (4.14.2)
Requirement already satisfied: h11>=0.8 in /opt/conda/lib/python3.12/site-packages (from uvicorn->sagemak
er<3,>=2) (0.16.0)
Requirement already satisfied: pycparser in /opt/conda/lib/python3.12/site-packages (from cffi>=1.14->cry
ptography<50,>=43.0.0->mlflow) (3.0)
Requirement already satisfied: mdurl~=0.1 in /opt/conda/lib/python3.12/site-packages (from markdown-it-py
>=2.2.0->rich<15.0.0,>=13.0.0->sagemaker-core<2.0.0,>=1.0.71->sagemaker<3,>=2) (0.1.2)
Requirement already satisfied: wcwidth>=0.3.5 in /opt/conda/lib/python3.12/site-packages (from prettytabl
e>=3.9->skops<1->mlflow) (0.8.2)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /opt/conda/lib/python3.12/site-packages (from pyas
n1-modules>=0.2.1->google-auth~=2.0->databricks-sdk<1,>=0.20.0->mlflow-skinny==3.15.1->mlflow) (0.6.4)
Collecting ppft>=1.7.8 (from pathos->sagemaker<3,>=2)
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Requirement already satisfied: dill>=0.4.1 in /opt/conda/lib/python3.12/site-packages (from pathos->sagem
aker<3,>=2) (0.4.1)
Collecting pox>=0.3.7 (from pathos->sagemaker<3,>=2)
  Using cached pox-0.3.7-py3-none-any.whl.metadata (8.0 kB)
Requirement already satisfied: multiprocess>=0.70.19 in /opt/conda/lib/python3.12/site-packages (from pat
hos->sagemaker<3,>=2) (0.70.19)
Using cached sagemaker-2.257.6-py3-none-any.whl (1.7 MB)
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Downloading boto3-1.43.78-py3-none-any.whl (140 kB)
Downloading botocore-1.43.78-py3-none-any.whl (15.7 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/15.7 MB ? eta -:--:--
   ━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.8/15.7 MB 5.1 MB/s eta 0:00:03
   ━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/15.7 MB 5.1 MB/s eta 0:00:03
   ━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.9/15.7 MB 5.1 MB/s eta 0:00:03
   ━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.9/15.7 MB 5.1 MB/s eta 0:00:03
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸ 15.5/15.7 MB 16.3 MB/s eta 0:00:01
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 15.7/15.7 MB 15.3 MB/s  0:00:01
Using cached attrs-25.4.0-py3-none-any.whl (67 kB)
Using cached packaging-24.2-py3-none-any.whl (65 kB)
Using cached sagemaker_core-1.0.78-py3-none-any.whl (444 kB)
Using cached mlflow-3.15.1-py3-none-any.whl (11.2 MB)
Using cached mlflow_skinny-3.15.1-py3-none-any.whl (3.6 MB)
Using cached mlflow_tracing-3.15.1-py3-none-any.whl (1.8 MB)
Using cached prettytable-3.18.0-py3-none-any.whl (37 kB)
Using cached pathos-0.3.5-py3-none-any.whl (82 kB)
Using cached pox-0.3.7-py3-none-any.whl (29 kB)
Using cached ppft-1.7.8-py3-none-any.whl (56 kB)
Installing collected packages: prettytable, ppft, pox, packaging, attrs, pathos, botocore, boto3, sagemak
er-core, mlflow-tracing, mlflow-skinny, sagemaker, mlflow
  Attempting uninstall: packaging

    Found existing installation: packaging 25.0

    Uninstalling packaging-25.0:

      Successfully uninstalled packaging-25.0
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
  Attempting uninstall: attrs
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
    Found existing installation: attrs 26.1.0
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
    Uninstalling attrs-26.1.0:
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
      Successfully uninstalled attrs-26.1.0
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
  Attempting uninstall: botocore
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
    Found existing installation: botocore 1.43.46
   ━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━  3/13 [packaging]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
    Uninstalling botocore-1.43.46:
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
      Successfully uninstalled botocore-1.43.46
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
  Attempting uninstall: boto3
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
    Found existing installation: boto3 1.43.46
   ━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━━━━  6/13 [botocore]
   ━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━  7/13 [boto3]
    Uninstalling boto3-1.43.46:
   ━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━  7/13 [boto3]
      Successfully uninstalled boto3-1.43.46
   ━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━  7/13 [boto3]
  Attempting uninstall: sagemaker-core
   ━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━  7/13 [boto3]
    Found existing installation: sagemaker-core 2.16.0
   ━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━━━━━━━  7/13 [boto3]
   ━━━━━━━━━━━━━━━━━━━━━━━━╸ ━━━━━━━━━━━━━━━  8/13 [sagemaker-core]
    Uninstalling sagemaker-core-2.16.0:
   ━━━━━━━━━━━━━━━━━━━━━━━━╸ ━━━━━━━━━━━━━━━  8/13 [sagemaker-core]
      Successfully uninstalled sagemaker-core-2.16.0
   ━━━━━━━━━━━━━━━━━━━━━━━━╸ ━━━━━━━━━━━━━━━  8/13 [sagemaker-core]
   ━━━━━━━━━━━━━━━━━━━━━━━━╸ ━━━━━━━━━━━━━━━  8/13 [sagemaker-core]
   ━━━━━━━━━━━━━━━━━━━━━━━━╸ ━━━━━━━━━━━━━━━  8/13 [sagemaker-core]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━  9/13 [mlflow-tracing]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━  9/13 [mlflow-tracing]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━  9/13 [mlflow-tracing]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━ ━━━━━━━━━━━━  9/13 [mlflow-tracing]
```

```
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
Attempting uninstall: mlflow-skinny
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
  Found existing installation: mlflow-skinny 3.13.0
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━━━━  9/13 [mlflow-tracing]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
  Uninstalling mlflow-skinny-3.13.0:
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
    Successfully uninstalled mlflow-skinny-3.13.0
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━━━━ 10/13 [mlflow-skinny]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
Attempting uninstall: mlflow
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
  Found existing installation: mlflow 3.13.0
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━━━━ 11/13 [sagemaker]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
  Uninstalling mlflow-3.13.0:
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
    Successfully uninstalled mlflow-3.13.0
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━╸━━━ 12/13 [mlflow]
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```

```
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ERROR: pip's dependency resolver does not currently take into account all the packages that are installe
d. This behaviour is the source of the following dependency conflicts.
autogluon-multimodal 1.5.0 requires nvidia-ml-py3<8.0,>=7.352.0, which is not installed.
autogluon-timeseries 1.5.0 requires chronos-forecasting<2.4,>=2.2.2, which is not installed.
autogluon-timeseries 1.5.0 requires einops<1,>=0.7, which is not installed.
autogluon-timeseries 1.5.0 requires peft<0.18,>=0.13.0, which is not installed.
aiobotocore 3.8.0 requires botocore<1.43.47,>=1.43.3, but you have botocore 1.43.78 which is incompatibl
e.
autogluon-common 1.5.0 requires pyarrow<21.0.0,>=7.0.0, but you have pyarrow 21.0.0 which is incompatibl
e.
autogluon-multimodal 1.5.0 requires fsspec[http]<=2025.3, but you have fsspec 2026.6.0 which is incompati
ble.
sagemaker-mlops 1.12.0 requires sagemaker-core>=2.12.0, but you have sagemaker-core 1.0.78 which is incom
patible.
sagemaker-serve 1.12.0 requires sagemaker-core>=2.12.0, but you have sagemaker-core 1.0.78 which is incom
patible.
sagemaker-studio-analytics-extension 0.3.0 requires sparkmagic==0.22.0, but you have sparkmagic 0.21.0 wh
ich is incompatible.
sagemaker-train 1.14.0 requires sagemaker-core>=2.14.0, but you have sagemaker-core 1.0.78 which is incom
patible.
sparkmagic 0.21.0 requires pandas<2.0.0,>=0.17.1, but you have pandas 2.3.3 which is incompatible.
Successfully installed attrs-25.4.0 boto3-1.43.78 botocore-1.43.78 mlflow-3.15.1 mlflow-skinny-3.15.1 mlf
low-tracing-3.15.1 packaging-24.2 pathos-0.3.5 pox-0.3.7 ppft-1.7.8 prettytable-3.18.0 sagemaker-2.257.6
sagemaker-core-1.0.78
Note: you may need to restart the kernel to use updated packages.
```

In [2]:

```python
import os
import json
import time
import boto3
import botocore
import numpy as np
import pandas as pd
import sagemaker
from sagemaker import get_execution_role
from sagemaker.workflow.pipeline_context import PipelineSession

print("SageMaker SDK:", sagemaker.__version__)
```

```
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/sagemaker-user/.config/sagemaker/c
onfig.yaml
/opt/conda/lib/python3.12/site-packages/sagemaker/__init__.py:86: SageMakerV2DeprecationWarning: You are
using the SageMaker Python SDK v2, which is on path of deprecation. v3 is the actively developed major ve
rsion.
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation()
You are using the SageMaker Python SDK v2, which is on path of deprecation. v3 is the actively developed
major version.
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
SageMaker SDK: 2.257.6
```

1. Configuration

In [3]:

```python
sess = boto3.Session()
role = get_execution_role()
sagemaker_session = sagemaker.Session(boto_session=sess)
pipeline_session = PipelineSession()
region = sess.region_name
bucket = sagemaker_session.default_bucket()



# Change these for the current student/profile.
TEAM_ID = "Team08"
STUDENT_ID = "s801"

COURSE = "ITI113"
SEMESTER = "26S1"
```

```python
PROJECT_NAME = "4697566D-Rental-Pipeline-V2"
PIPELINE_NAME = PROJECT_NAME
MODEL_PACKAGE_GROUP = "RentalPriceModelGroupV2"
BASE_PREFIX = "sgp-rental-v2"

TEAM_ID = "Team08"
STUDENT_ID = "s801"

COURSE = "ITI113"
SEMESTER = "26S1"

PROCESSING_INSTANCE = "ml.m5.large"
TRAINING_INSTANCE = "ml.m5.large"
INFERENCE_INSTANCE = "ml.m5.large"

print("Region:", region)
print("Bucket:", bucket)
print("Role:", role)
print("Model group:", MODEL_PACKAGE_GROUP)

# ----------------------------
# MLflow App UI link helpers
# ----------------------------
# MLflow may print generic links such as:
# https://mlflow.sagemaker.ap-southeast-1.app.aws/#/...
# Those links are not presigned and may show a permission/session error.
# Use these helpers to generate a fresh presigned SageMaker MLflow App URL
# and append the experiment/run fragment.
MLFLOW_EXPERIMENT_NAME = f"{COURSE}/{TEAM_ID}/Experiment"

def create_mlflow_app_presigned_url(fragment: str = "") -> str:
    sm_for_mlflow = boto3.client("sagemaker", region_name=region)
    response = sm_for_mlflow.create_presigned_mlflow_app_url(
        Arn=MLFLOW_APP_ARN
    )

    base_url = response.get("AuthorizedUrl") or response.get("Url")

    if not base_url:
        raise RuntimeError(
            "create_presigned_mlflow_app_url did not return AuthorizedUrl or Url. "
            f"Response: {response}"
        )

    # Remove any existing fragment before appending our own MLflow UI route.
    base_url = base_url.split("#", 1)[0]

    if fragment:
        return base_url + "#" + fragment.lstrip("#")

    return base_url


def print_mlflow_presigned_links(experiment_id=None, run_id=None):
    if experiment_id is not None:
        experiment_url = create_mlflow_app_presigned_url(
            f"/experiments/{experiment_id}"
        )
        print("Presigned MLflow experiment URL:")
        print(experiment_url)

    if experiment_id is not None and run_id is not None:
        run_url = create_mlflow_app_presigned_url(
            f"/experiments/{experiment_id}/runs/{run_id}"
        )
        print("\nPresigned MLflow run URL:")
        print(run_url)
```

```
Region: ap-southeast-1
Bucket: sagemaker-ap-southeast-1-044528205969
Role: arn:aws:iam::044528205969:role/SageMakerExecutionRole-ITI113-Team08
Model group: RentalPriceModelGroupV2
```

In [4]:

```python
import boto3
import time
```

```python
import json
from botocore.exceptions import ClientError

MLFLOW_APP_NAME = "iti113-26s1-team08-mlflow-app"


REGION = "ap-southeast-1"
MLFLOW_APP_ARN = "arn:aws:sagemaker:ap-southeast-1:044528205969:mlflow-app/app-WLHEEZL7YYAH"

sm = boto3.client("sagemaker", region_name=REGION)

while True:
    try:
        response = sm.describe_mlflow_app(
            Arn=MLFLOW_APP_ARN
        )

        print("=" * 80)
        print("Name:", response.get("Name"))
        print("Status:", response.get("Status"))
        print("Arn:", response.get("Arn"))
        print("LastModifiedTime:", response.get("LastModifiedTime"))

        if response.get("Status") in ["Created", "Failed", "Deleting", "Deleted"]:
            break

    except ClientError as e:
        print("Describe failed.")
        print(e.response["Error"]["Code"])
        print(e.response["Error"]["Message"])
        break

    time.sleep(30)
```

```
================================================================================
Name: iti113-26s1-team08-mlflow-app
Status: Created
Arn: arn:aws:sagemaker:ap-southeast-1:044528205969:mlflow-app/app-WLHEEZL7YYAH
LastModifiedTime: 2026-08-19 13:01:50.294000+00:00
```

In [4]:

```python
import boto3
import time
import json
from botocore.exceptions import ClientError


REGION = "ap-southeast-1"
ACCOUNT_ID = "044528205969"
MLFLOW_APP_ARN = "arn:aws:sagemaker:ap-southeast-1:044528205969:mlflow-app/app-WLHEEZL7YYAH"

TEAM_ID = "team08"
STUDENT_ID = "s801"

MLFLOW_APP_NAME = "iti113-26s1-team08-mlflow-app"
ARTIFACT_STORE_URI = "s3://nyp-26s1-iti113/iti113/team08/mlflow-app-artifacts/"
ROLE_ARN = f"arn:aws:iam::{ACCOUNT_ID}:role/SageMakerExecutionRole-ITI113-Team08"

sm = boto3.client("sagemaker", region_name=REGION)

while True:
    try:
        response = sm.describe_mlflow_app(
            Arn=MLFLOW_APP_ARN
        )

        print("=" * 80)
        print("Name:", response.get("Name"))
        print("Status:", response.get("Status"))
        print("Arn:", response.get("Arn"))
        print("LastModifiedTime:", response.get("LastModifiedTime"))

        if response.get("Status") in ["Created", "Failed", "Deleting", "Deleted"]:
            break

    except ClientError as e:
```

```
        print("Describe failed.")
        print(e.response["Error"]["Code"])
        print(e.response["Error"]["Message"])
        break

    time.sleep(30)
```

```
================================================================================
Name: iti113-26s1-team08-mlflow-app
Status: Created
Arn: arn:aws:sagemaker:ap-southeast-1:044528205969:mlflow-app/app-WLHEEZL7YYAH
LastModifiedTime: 2026-08-19 13:01:50.294000+00:00
```

2. Prepare and upload the cleaned dataset##

In [5]:

```
import numpy as np
import pandas as pd

#upload to Jupyter Server

rental = pd.read_csv("rent21_25_yr.csv", index_col=0)

rental.rename(columns={
    "Project Name": "project",
    "Street Name": "street",
    "Postal District": "district",
    "Property Type": "type",
    "No of Bedroom": "bedroom",
    "Monthly Rent ($)": "mth rent",
    "Floor Area (SQM)": "area_sm",
    "Floor Area (SQFT)": "area_sf",
    "Lease Commencement Date": "lease date"
}, inplace=True)

rental_cl = rental.dropna().copy()
rental_cl.drop(columns=["type", "area_sf"], errors="ignore", inplace=True)
rental_cl["lease date"] = pd.to_datetime(rental_cl["lease date"], format="%b-%y")
rental_cl["lease_year"] = rental_cl["lease date"].dt.year
rental_cl["lease_month"] = rental_cl["lease date"].dt.month
rental_cl.drop(columns=["lease date"], inplace=True)

rental_cl["area_sm"] = (
    rental_cl["area_sm"].astype(str).str.split("to", expand=True)[1].str.strip()
)

rental_cl = rental_cl[
    pd.to_numeric(rental_cl["district"], errors="coerce").notnull() &
    pd.to_numeric(rental_cl["bedroom"], errors="coerce").notnull() &
    pd.to_numeric(rental_cl["area_sm"], errors="coerce").notnull()
].copy()

rental_cl["district"] = rental_cl["district"].astype(int)
rental_cl["bedroom"] = rental_cl["bedroom"].astype(int)
rental_cl["area_sm"] = rental_cl["area_sm"].astype(int)
rental_cl["mth rent"] = rental_cl["mth rent"].astype(str).str.replace(",", "", regex=False)
rental_cl = rental_cl[pd.to_numeric(rental_cl["mth rent"], errors="coerce").notnull()].copy()
rental_cl["mth rent"] = (rental_cl["mth rent"].astype(float) / 1000).round(3)

rental_2125 = rental_cl.copy()
rental_2125["price_per_sqm"] = (rental_2125["mth rent"] * 1000 / rental_2125["area_sm"]).round(3)

rental_2125.to_csv("rental_2125.csv", index=False)
print("Rows:", len(rental_2125))
print("Columns:", list(rental_2125.columns))

input_data_uri = sagemaker_session.upload_data(
    path="rental_2125.csv",
    bucket=bucket,
    key_prefix=f"{BASE_PREFIX}/input"
)
print("Input S3:", input_data_uri)
```

```
Rows: 368897
Columns: ['project', 'street', 'district', 'bedroom', 'mth rent', 'area_sm', 'lease_year', 'lease_month', 'price_per_sqm']
Input S3: s3://sagemaker-ap-southeast-1-044528205969/sgp-rental-v2/input/rental_2125.csv
```

In [6]:
```
## 3. Pipeline parameters
```

In [7]:
```
from sagemaker.workflow.parameters import ParameterString, ParameterFloat

input_data = ParameterString(name="InputData", default_value=input_data_uri)
processing_instance_type = ParameterString(
    name="ProcessingInstanceType", default_value=PROCESSING_INSTANCE
)
training_instance_type = ParameterString(
    name="TrainingInstanceType", default_value=TRAINING_INSTANCE
)
model_approval_status = ParameterString(
    name="ModelApprovalStatus", default_value="Approved"
)
r2_threshold = ParameterFloat(name="R2Threshold", default_value=0.8)
```

4. Create the source directory

In [8]:
```
!mkdir -p src
```

5. Preprocessing script

In [9]:
```
%%writefile src/preprocess.py
import pandas as pd
import numpy as np
import os
from sklearn.preprocessing import OneHotEncoder

if __name__ == "__main__":
    input_dir = "/opt/ml/processing/input"
    input_file = [f for f in os.listdir(input_dir) if f.endswith(".csv")][0]
    df = pd.read_csv(os.path.join(input_dir, input_file))

    dist_dummy_cols = [c for c in df.columns if c.startswith("dist_")]
    columns_to_drop = ["project", "street","price_per_sqm"] + dist_dummy_cols
    df_cleaned = df.drop(columns=columns_to_drop, errors="ignore")


    target_col = "mth rent"
    shuffled = df_cleaned.sample(frac=1, random_state=42).reset_index(drop=True)
    n = len(shuffled)
    train_end = int(0.70 * n)
    val_end = int(0.85 * n)
    train_df = shuffled.iloc[:train_end]
    val_df = shuffled.iloc[train_end:val_end]
    test_df = shuffled.iloc[val_end:]

    encoder = OneHotEncoder(handle_unknown="ignore", sparse_output=False)
    encoder.fit(train_df[["district"]])
    district_cols = [f"dist_{int(c)}" for c in encoder.categories_[0]]

    def build_output(split_df):
        district_encoded = pd.DataFrame(
            encoder.transform(split_df[["district"]]).astype(int),
            columns=district_cols,
            index=split_df.index,
        )
        numeric_cols = ["bedroom", "area_sm", "lease_year", "lease_month"]
        return pd.concat(
            [split_df[[target_col] + numeric_cols].reset_index(drop=True),
             district_encoded.reset_index(drop=True)], axis=1
        )

    train_out = build_output(train_df)
    val_out = build_output(val_df)
    test_out = build_output(test_df)
```

```
    for d in ["/opt/ml/processing/train", "/opt/ml/processing/validation", "/opt/ml/processing/test"]:
        os.makedirs(d, exist_ok=True)

    train_out.to_csv("/opt/ml/processing/train/train.csv", header=False, index=False)
    val_out.to_csv("/opt/ml/processing/validation/validation.csv", header=False, index=False)
    test_out.to_csv("/opt/ml/processing/test/test.csv", header=False, index=False)

    print(f"Preprocessing finished: train={len(train_out)}, validation={len(val_out)}, test={len(test_ou
    print("Feature columns:", list(train_out.columns[1:]))
```

```
Overwriting src/preprocess.py
```

In [10]:

```
from sagemaker.processing import ScriptProcessor, ProcessingInput, ProcessingOutput
from sagemaker.workflow.steps import ProcessingStep

sklearn_image = sagemaker.image_uris.retrieve(
    framework="sklearn", region=region, version="1.2-1"
)

processor = ScriptProcessor(
    image_uri=sklearn_image,
    command=["python3"],
    instance_type=processing_instance_type,
    instance_count=1,
    role=role,
    sagemaker_session=pipeline_session,
)

step_process = ProcessingStep(
    name="PreprocessData",
    processor=processor,
    inputs=[ProcessingInput(source=input_data, destination="/opt/ml/processing/input")],
    outputs=[
        ProcessingOutput(output_name="train", source="/opt/ml/processing/train"),
        ProcessingOutput(output_name="validation", source="/opt/ml/processing/validation"),
        ProcessingOutput(output_name="test", source="/opt/ml/processing/test"),
    ],
    code="src/preprocess.py",
)
```

```
/opt/conda/lib/python3.12/site-packages/sagemaker/processing.py:138: SageMakerV2DeprecationWarning: Scrip
tProcessor is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `DataProce
ssor` (`from sagemaker.mlops.processing import DataProcessor`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
ScriptProcessor is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `Data
Processor` (`from sagemaker.mlops.processing import DataProcessor`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
```

6. XGBoost training script

In [11]:

```
import mlflow
print("MLflow version:", mlflow.__version__)
```

```
MLflow version: 3.15.1
```

In [12]:

```
%%writefile src/train.py
import argparse
import os
import sys
import subprocess

# The SageMaker XGBoost container does not necessarily include the
# SageMaker MLflow integration packages, so install them in the training
# container before importing MLflow.
subprocess.check_call(
    [sys.executable, "-m", "pip", "install", "-q", "sagemaker-mlflow"]
)

import pandas as pd
import xgboost as xgb
```

```python
import numpy as np
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error

import mlflow
import mlflow.xgboost


def configure_mlflow():
    """Connect this training job to the SageMaker MLflow App."""
    mlflow_app_arn = os.environ.get("MLFLOW_APP_ARN")
    experiment_name = os.environ.get(
        "MLFLOW_EXPERIMENT_NAME",
        "4697566D-Rental-Pipeline-V2-MLflow",
    )

    if not mlflow_app_arn:
        raise RuntimeError(
            "MLFLOW_APP_ARN is not set. The SageMaker training step "
            "must pass the MLflow App ARN in the environment."
        )

    # IMPORTANT:
    # This ARN is an MLflow App ARN:
    # arn:aws:sagemaker:<region>:<account>:mlflow-app/<id>
    # It is NOT an MLflow Tracking Server ARN.
    mlflow.set_tracking_uri(mlflow_app_arn)
    mlflow.set_experiment(experiment_name)

    return experiment_name


if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--eta", type=float, default=0.2)
    parser.add_argument("--max_depth", type=int, default=6)
    parser.add_argument("--num_round", type=int, default=400)
    parser.add_argument("--objective", type=str, default="reg:squarederror")
    parser.add_argument("--subsample", type=float, default=0.8)
    parser.add_argument("--eval_metric", type=str, default="rmse")
    parser.add_argument("--train", type=str, default=os.environ.get("SM_CHANNEL_TRAIN"))
    parser.add_argument("--validation", type=str, default=os.environ.get("SM_CHANNEL_VALIDATION"))
    parser.add_argument("--model-dir", type=str, default=os.environ.get("SM_MODEL_DIR", "/opt/ml/model")
    args, _ = parser.parse_known_args()

    train_df = pd.read_csv(
        os.path.join(args.train, "train.csv"), header=None
    )
    val_df = pd.read_csv(
        os.path.join(args.validation, "validation.csv"), header=None
    )

    x_train, y_train = train_df.iloc[:, 1:], train_df.iloc[:, 0]
    x_val, y_val = val_df.iloc[:, 1:], val_df.iloc[:, 0]

    model = xgb.XGBRegressor(
        n_estimators=args.num_round,
        learning_rate=args.eta,
        max_depth=args.max_depth,
        subsample=args.subsample,
        objective=args.objective,
        eval_metric=args.eval_metric,
        random_state=42,
        verbosity=1,
    )

    # ----------------------------------------------------------------------
    # MLflow tracking
    # ----------------------------------------------------------------------
    experiment_name = configure_mlflow()

    training_job_name = os.environ.get(
        "TRAINING_JOB_NAME",
        os.environ.get("SAGEMAKER_JOB_NAME", "rental-xgboost-training"),
    )
```

```python
    with mlflow.start_run(run_name=training_job_name) as run:

        # Record the hyperparameters used by the XGBoost model.
        mlflow.log_params({
            "eta": args.eta,
            "max_depth": args.max_depth,
            "num_round": args.num_round,
            "objective": args.objective,
            "subsample": args.subsample,
            "eval_metric": args.eval_metric,
            "random_state": 42,
        })

        # Record useful dataset information.
        mlflow.log_params({
            "training_samples": len(train_df),
            "validation_samples": len(val_df),
            "features": x_train.shape[1],
        })

        mlflow.set_tags({
            "project": "4697566D My Rental Price",
            "pipeline": "4697566D-Rental-Pipeline-V2",
            "framework": "XGBoost",
            "sagemaker_training_job": training_job_name,
        })

        # Train the model.
        model.fit(x_train, y_train)

        # Validate the model.
        pred = model.predict(x_val)

        rmse = float(np.sqrt(mean_squared_error(y_val, pred)))
        mae = float(mean_absolute_error(y_val, pred))
        r2 = float(r2_score(y_val, pred))

        # Record evaluation metrics in MLflow.
        mlflow.log_metrics({
            "rmse": rmse,
            "mae": mae,
            "r2": r2,
        })

        # Save the same SageMaker model artifact used by the existing
        # RegisterModel/deployment steps.
        os.makedirs(args.model_dir, exist_ok=True)

        model.save_model(
            os.path.join(args.model_dir, "xgboost-model.ubj")
        )
        model.save_model(
            os.path.join(args.model_dir, "xgboost-model.json")
        )

        # Also record the trained model in the MLflow App.
        mlflow.xgboost.log_model(
            model,
            name="xgboost-model",
            registered_model_name="ITI113-Team08-SGP-Rental-Price-Model",
        )

        print("MLflow experiment:", experiment_name)
        print("MLflow run ID:", run.info.run_id)

    print(f"RMSE={rmse:.4f}")
    print(f"MAE={mae:.4f}")
    print(f"R2={r2:.4f}")
    print(f"Training samples={len(train_df)}")
    print(f"Validation samples={len(val_df)}")
    print(f"Features={x_train.shape[1]}")
```

```
Overwriting src/train.py
```

7. Training step

In [13]:

```python
from sagemaker.xgboost.estimator import XGBoost
from sagemaker.inputs import TrainingInput
from sagemaker.workflow.steps import TrainingStep

xgb_estimator = XGBoost(
    entry_point="train.py",
    source_dir="src",
    framework_version="1.7-1",
    py_version="py3",
    role=role,
    instance_type=training_instance_type,
    instance_count=1,
    hyperparameters={
        "max_depth": 6,
        "eta": 0.2,
        "subsample": 0.8,
        "objective": "reg:squarederror",
        "num_round": 400,
        "eval_metric": "rmse",
    },

    # Pass the SageMaker MLflow App information into the training container.
    # train.py uses this ARN as its MLflow tracking URI.
    environment={
        "MLFLOW_APP_ARN": MLFLOW_APP_ARN,
        "MLFLOW_EXPERIMENT_NAME": MLFLOW_EXPERIMENT_NAME,
    },

    sagemaker_session=pipeline_session,
)


# `xgboost` library

assert isinstance(xgb_estimator, XGBoost), (
    f"xgb_estimator is a {type(xgb_estimator)}, expected "
    f"sagemaker.xgboost.estimator.XGBoost. This usually means the "
    f"kernel has stale state or a name collision with the local "
    f"xgboost library -- restart the kernel and run all cells in order."
)

step_train = TrainingStep(
    name="TrainXGBoostModel",
    estimator=xgb_estimator,
    inputs={
        "train": TrainingInput(
            s3_data=step_process.properties.ProcessingOutputConfig.Outputs["train"].S3Output.S3Uri,
            content_type="text/csv",
        ),
        "validation": TrainingInput(
            s3_data=step_process.properties.ProcessingOutputConfig.Outputs["validation"].S3Output.S3Uri,
            content_type="text/csv",
        ),
    },
)
```

```
/opt/conda/lib/python3.12/site-packages/sagemaker/estimator.py:588: SageMakerV2DeprecationWarning: XGBoos
t is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelTrainer` (`fr
om sagemaker.train import ModelTrainer`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
XGBoost is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelTrainer
` (`from sagemaker.train import ModelTrainer`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
The input argument instance_type of function (sagemaker.image_uris.retrieve) is a pipeline variable (<cla
ss 'sagemaker.workflow.parameters.ParameterString'>), which is interpreted in pipeline execution time onl
y. As the function needs to evaluate the argument value in SDK compile time, the default_value of this Pa
rameter object will be used to override it. Please make sure the default_value is valid.
```

9. Evaluation step

In [15]:
```
%%writefile src/evaluation.py
import json
import os
import tarfile
import numpy as np
import pandas as pd
import xgboost as xgb
from sklearn.metrics import mean_squared_error, r2_score

model_dir = "/opt/ml/processing/model"
with tarfile.open(os.path.join(model_dir, "model.tar.gz")) as tar:
    tar.extractall(path=model_dir)

model = xgb.XGBRegressor()
model_path = os.path.join(model_dir, "xgboost-model.ubj")
if not os.path.exists(model_path):
    model_path = os.path.join(model_dir, "xgboost-model.json")
model.load_model(model_path)

test_dir = "/opt/ml/processing/test"
test_file = [f for f in os.listdir(test_dir) if f.endswith(".csv")][0]
test_df = pd.read_csv(os.path.join(test_dir, test_file), header=None)
y_test = test_df.iloc[:, 0]
x_test = test_df.iloc[:, 1:]
predictions = model.predict(x_test)

rmse = float(np.sqrt(mean_squared_error(y_test, predictions)))
r2 = float(r2_score(y_test, predictions))
report = {"regression_metrics": {"rmse": {"value": rmse, "standard_deviation": "NaN"}, "r2": {"value": r

output_dir = "/opt/ml/processing/evaluation"
os.makedirs(output_dir, exist_ok=True)
with open(os.path.join(output_dir, "evaluation.json"), "w") as f:
    json.dump(report, f)
print(f"RMSE={rmse:.4f} R2={r2:.4f}")
```

```
Overwriting src/evaluation.py
```

In [14]:
```
from sagemaker.processing import ScriptProcessor, ProcessingInput, ProcessingOutput
from sagemaker.workflow.properties import PropertyFile
from sagemaker.workflow.steps import ProcessingStep

xgb_image = sagemaker.image_uris.retrieve(
    framework="xgboost",
    region=region,
    version="1.7-1"
)


eval_processor = ScriptProcessor(
    image_uri=xgb_image, command=["python3"],
    instance_type=processing_instance_type, instance_count=1,
    role=role, sagemaker_session=pipeline_session
)

evaluation_report = PropertyFile(name="EvaluationReport", output_name="evaluation", path="evaluation.jso

step_eval = ProcessingStep(
    name="EvaluateModel",
    processor=eval_processor,
    inputs=[
        ProcessingInput(source=step_train.properties.ModelArtifacts.S3ModelArtifacts, destination="/opt/
        ProcessingInput(source=step_process.properties.ProcessingOutputConfig.Outputs["test"].S3Output.S
    ],
    outputs=[ProcessingOutput(output_name="evaluation", source="/opt/ml/processing/evaluation")],
    code="src/evaluation.py",
    property_files=[evaluation_report],
)
```

8. Adding Custom inference script to accept both JSON + CSV input with the model

In [15]:
```python
%%writefile src/inference.py

#Custom SageMaker inference script for the Singapore Rent Price XGBoost model.

import os
import json
from io import StringIO

import pandas as pd
import xgboost as xgb



# DISTRICT CATEGORIES
# Singapore postal districts present in the training data.
# (24 is absent — not a valid postal district.)

DISTRICTS = [
    1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
    11, 12, 13, 14, 15, 16, 17, 18, 19,
    20, 21, 22, 23, 25, 26, 27, 28,
]


# LOGICAL FEATURE NAMES
#[target] + [bedroom, area_sm, lease_year, lease_month] + dist_1..dist_28


FEATURE_NAMES = [
    "bedroom",
    "area_sm",
    "lease_year",
    "lease_month",
] + [f"dist_{d}" for d in DISTRICTS]

EXPECTED_FEATURE_COUNT = len(FEATURE_NAMES)  # 31 columns



# MODEL LOADING


def model_fn(model_dir):
    # train.py saves both text/csv and .json is the portable/preferred format
    model_path = os.path.join(model_dir, "xgboost-model.json")

    print("Loading XGBoost model")
    print("Model directory:", model_dir)
    print("Model path:", model_path)
    print("Files in model directory:", end=" ")
    try:
        print(os.listdir(model_dir))
    except Exception as e:
        print(f"(unable to list: {e})")

    if not os.path.exists(model_path):
        # Fall back to the .ubj artifact if .json isn't present for any reason.
        fallback_path = os.path.join(model_dir, "xgboost-model.ubj")
        if os.path.exists(fallback_path):
            model_path = fallback_path
        else:
            raise FileNotFoundError(
                f"Model file not found at {model_path} or {fallback_path}. "
                f"Contents of model_dir: {os.listdir(model_dir)}"
            )

    model = xgb.XGBRegressor()
    model.load_model(model_path)
    print("XGBoost model loaded successfully")

    booster = model.get_booster()
    print("Number of model features:", booster.num_features())
    print("Model feature names:", booster.feature_names)
```

```
    if booster.num_features() != EXPECTED_FEATURE_COUNT:
        raise ValueError(
            f"Expected model to have {EXPECTED_FEATURE_COUNT} features, "
            f"but model has {booster.num_features()}"
        )

    return model

# NORMALISE A JSON RECORD
# Accepts either the fully expanded one-hot form, or the compact
# form with a single "district" key that gets expanded here.

def _normalise_record(record):
    if not isinstance(record, dict):
        raise ValueError("Each JSON record must be an object.")

    record = dict(record)

    if "district" in record:
        district = int(record.pop("district"))
        if district not in DISTRICTS:
            raise ValueError(
                f"Invalid district {district}. Valid districts are {DISTRICTS}"
            )
        for d in DISTRICTS:
            record[f"dist_{d}"] = 1 if d == district else 0

    return record



# VALIDATE INPUT
# bedroom and area_sm must be whole numbers.


def _validate_whole_numbers(df):
    for col in ("bedroom", "area_sm"):
        if col in df.columns:
            values = pd.to_numeric(df[col], errors="raise")
            if (values != values.round(0)).any():
                raise ValueError(f"{col} must contain whole numbers only")



# INPUT FUNCTION
# Supports application/json and text/csv input


def input_fn(request_body, content_type):

    print("INPUT REQUEST")

    print("Content type:", content_type)

    if content_type == "application/json":
        if isinstance(request_body, bytes):
            request_body = request_body.decode("utf-8")

        data = json.loads(request_body)

        if isinstance(data, dict):
            records = [_normalise_record(data)]
        elif isinstance(data, list):
            records = [_normalise_record(r) for r in data]
        else:
            raise ValueError("JSON must be an object or a list of objects.")

        df = pd.DataFrame(records)

        # Fill any omitted feature columns unselected districts) with 0
        for column in FEATURE_NAMES:
            if column not in df.columns:
                df[column] = 0

        # Enforce the exact column order the model was trained on
```

```
        df = df[FEATURE_NAMES]
        _validate_whole_numbers(df)

        print("JSON dataframe shape:", df.shape)
        print("Logical feature columns:", list(df.columns))
        return df

    if content_type == "text/csv":
        if isinstance(request_body, bytes):
            request_body = request_body.decode("utf-8")

        df = pd.read_csv(StringIO(request_body), header=None)

        if df.shape[1] == 5:
            # Compact format, matching the JSON input's shape:
            #  bedroom, area_sm, lease_year, lease_month, district
            # district is expanded into the one-hot dist_* columns here,
            # so callers don't have to build all 31 values by hand.
            df.columns = ["bedroom", "area_sm", "lease_year", "lease_month", "district"]

            district_col = pd.to_numeric(df["district"], errors="raise").astype(int)
            invalid = ~district_col.isin(DISTRICTS)
            if invalid.any():
                raise ValueError(
                    f"Invalid district value(s): {district_col[invalid].tolist()}. "
                    f"Valid districts are {DISTRICTS}"
                )

            for d in DISTRICTS:
                df[f"dist_{d}"] = (district_col == d).astype(int)

            df = df.drop(columns=["district"])
            df = df[FEATURE_NAMES]

        elif df.shape[1] == EXPECTED_FEATURE_COUNT:
            # Fully expanded format: 4 numeric features + 27 one-hot
            df.columns = FEATURE_NAMES

        else:
            raise ValueError(
                f"CSV must contain either 5 columns (bedroom, area_sm, "
                f"lease_year, lease_month, district) or "
                f"{EXPECTED_FEATURE_COUNT} fully expanded feature columns. "
                f"Received {df.shape[1]}."
            )

        _validate_whole_numbers(df)

        print("CSV dataframe shape:", df.shape)
        print("Logical feature columns:", list(df.columns))
        return df

    raise ValueError(
        f"Unsupported content type: {content_type}. "
        f"Supported types are 'application/json' and 'text/csv'."
    )



# PREDICTION


def predict_fn(input_data, model):

    print("PREDICTION")

    input_data = input_data.copy()

    if input_data.shape[1] != EXPECTED_FEATURE_COUNT:
        raise ValueError(
            f"Expected {EXPECTED_FEATURE_COUNT} input features, "
            f"received {input_data.shape[1]}"
        )

    # column labels train.py's DataFrame had at fit time (headerless CSV,
```

```
    # columns 1..31 after dropping the target column)
    input_data.columns = [str(i) for i in range(1, EXPECTED_FEATURE_COUNT + 1)]

    print("Columns passed to XGBoost:", list(input_data.columns))
    print("Input shape:", input_data.shape)

    prediction = model.predict(input_data)
    print("Raw prediction (thousands scale):", prediction)
    return prediction



# OUTPUT FUNCTION
# Model was trained on rent scaled to thousands (mth_rent / 1000),


def output_fn(prediction, accept):

    print("OUTPUT")

    # Convert prediction back from thousands to dollars
    predictions = [round(float(p) * 1000) for p in prediction.tolist()]

    print("Prediction values ($):", predictions)

    if accept == "text/csv":
        return ",".join(str(p) for p in predictions), "text/csv"

    return json.dumps(predictions), "application/json"
```

```
Overwriting src/inference.py
```

SageMaker `Model` using the training artifact and `src/inference.py`, then register that model. This preserves the custom serving code in the model package.

In [16]:

```
from sagemaker.xgboost.model import XGBoostModel
from sagemaker.model_metrics import ModelMetrics, MetricsSource
from sagemaker.workflow.step_collections import RegisterModel
from sagemaker.workflow.condition_step import ConditionStep
from sagemaker.workflow.conditions import ConditionGreaterThanOrEqualTo
from sagemaker.workflow.functions import JsonGet, Join
from sagemaker.workflow.fail_step import FailStep



# XGBoost serving image

xgb_image = sagemaker.image_uris.retrieve(
    framework="xgboost",
    region=region,
    version="1.7-1"
)



# Framework-specific XGBoost model


rent_model = XGBoostModel(
    image_uri=xgb_image,
    model_data=step_train.properties.ModelArtifacts.S3ModelArtifacts,
    role=role,

    # Custom inference code
    entry_point="inference.py",
    source_dir="src",

    framework_version="1.7-1",

    sagemaker_session=pipeline_session,
)
```

```python
# Evaluation metrics


model_metrics = ModelMetrics(
    model_statistics=MetricsSource(
        s3_uri=Join(
            on="/",
            values=[
                step_eval.properties
                    .ProcessingOutputConfig
                    .Outputs["evaluation"]
                    .S3Output
                    .S3Uri,
                "evaluation.json",
            ],
        ),
        content_type="application/json",
    )
)



# Register model


step_register = RegisterModel(
    name="RegisterModelV2",

    model=rent_model,

    content_types=[
        "application/json",
        "text/csv",
    ],

    response_types=[
        "application/json",
        "text/csv",
    ],

    inference_instances=[
        INFERENCE_INSTANCE
    ],

    transform_instances=[
        INFERENCE_INSTANCE
    ],

    model_package_group_name=MODEL_PACKAGE_GROUP,

    approval_status=model_approval_status,

    model_metrics=model_metrics,
)



# Fail step


step_fail = FailStep(
    name="ModelQualityFailed",
    error_message="Model R2 below threshold.",
)


# R2 quality gate

step_cond = ConditionStep(
    name="CheckR2Threshold",

    conditions=[
        ConditionGreaterThanOrEqualTo(
            left=JsonGet(
                step_name=step_eval.name,
```

```
                property_file=evaluation_report,
                json_path="regression_metrics.r2.value",
            ),
            right=r2_threshold,
        )
    ],

    if_steps=[
        step_register
    ],

    else_steps=[
        step_fail
    ],
)
```

```
/opt/conda/lib/python3.12/site-packages/sagemaker/model.py:347: SageMakerV2DeprecationWarning: XGBoostMod
el is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelBuilder` (`f
rom sagemaker.serve import ModelBuilder`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
XGBoostModel is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelBu
ilder` (`from sagemaker.serve import ModelBuilder`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
/opt/conda/lib/python3.12/site-packages/sagemaker/estimator.py:588: SageMakerV2DeprecationWarning: SKLear
n is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelTrainer` (`fr
om sagemaker.train import ModelTrainer`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
SKLearn is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelTrainer
` (`from sagemaker.train import ModelTrainer`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
```

12. Build and run the pipeline

In [17]:

```
from sagemaker.workflow.pipeline import Pipeline

pipeline = Pipeline(
    name=PIPELINE_NAME,
    parameters=[input_data, processing_instance_type, training_instance_type, model_approval_status, r2_
    steps=[step_process, step_train, step_eval, step_cond],
    sagemaker_session=pipeline_session,
)

pipeline.upsert(role_arn=role)
print("Pipeline upserted:", PIPELINE_NAME)

execution = pipeline.start()
print("Execution started:", execution.arn)
```

```
/opt/conda/lib/python3.12/site-packages/sagemaker/workflow/pipeline.py:119: SageMakerV2DeprecationWarnin
g: Pipeline is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `Pipeline
` (`from sagemaker.mlops.pipeline import Pipeline`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
WARNING:sagemaker.deprecations:Pipeline is part of the SageMaker Python SDK v2, which is on path of depre
cation. In v3, use `Pipeline` (`from sagemaker.mlops.pipeline import Pipeline`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
INFO:sagemaker.telemetry.telemetry_logging:SageMaker Python SDK will collect telemetry to help us better
understand our user's needs, diagnose issues, and deliver additional features.
To opt out of telemetry, please disable via TelemetryOptOut parameter in SDK defaults config. For more in
formation, refer to https://sagemaker.readthedocs.io/en/stable/overview.html#configuring-and-using-defaul
ts-with-the-sagemaker-python-sdk.
WARNING:sagemaker.workflow.utilities:Popping out 'ProcessingJobName' from the pipeline definition by defa
ult since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig t
o persist this field in the pipeline definition if desired.
```

```
WARNING:sagemaker.workflow.utilities:The input argument instance_type of function (sagemaker.image_uris.get_training_image_uri) is a pipeline variable (<class 'sagemaker.workflow.parameters.ParameterString'>), which is interpreted in pipeline execution time only. As the function needs to evaluate the argument value in SDK compile time, the default_value of this Parameter object will be used to override it. Please make sure the default_value is valid.
WARNING:sagemaker.workflow.utilities:The input argument instance_type of function (sagemaker.image_uris.get_training_image_uri) is a pipeline variable (<class 'sagemaker.workflow.parameters.ParameterString'>), which is interpreted in pipeline execution time only. As the function needs to evaluate the argument value in SDK compile time, the default_value of this Parameter object will be used to override it. Please make sure the default_value is valid.
WARNING:sagemaker.workflow.utilities:Popping out 'TrainingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:Popping out 'ProcessingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:Popping out 'TrainingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow._utils:Popping out 'CertifyForMarketplace' from the pipeline definition since it will be overridden in pipeline execution time.
WARNING:sagemaker.workflow.utilities:Popping out 'ModelPackageName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:Popping out 'ProcessingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:The input argument instance_type of function (sagemaker.image_uris.get_training_image_uri) is a pipeline variable (<class 'sagemaker.workflow.parameters.ParameterString'>), which is interpreted in pipeline execution time only. As the function needs to evaluate the argument value in SDK compile time, the default_value of this Parameter object will be used to override it. Please make sure the default_value is valid.
WARNING:sagemaker.workflow.utilities:Popping out 'TrainingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:Popping out 'ProcessingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow.utilities:Popping out 'TrainingJobName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
WARNING:sagemaker.workflow._utils:Popping out 'CertifyForMarketplace' from the pipeline definition since it will be overridden in pipeline execution time.
WARNING:sagemaker.workflow.utilities:Popping out 'ModelPackageName' from the pipeline definition by default since it will be overridden at pipeline execution time. Please utilize the PipelineDefinitionConfig to persist this field in the pipeline definition if desired.
INFO:sagemaker.telemetry.telemetry_logging:SageMaker Python SDK will collect telemetry to help us better understand our user's needs, diagnose issues, and deliver additional features.
To opt out of telemetry, please disable via TelemetryOptOut parameter in SDK defaults config. For more information, refer to https://sagemaker.readthedocs.io/en/stable/overview.html#configuring-and-using-defaults-with-the-sagemaker-python-sdk.
Pipeline upserted: 4697566D-Rental-Pipeline-V2
Execution started: arn:aws:sagemaker:ap-southeast-1:044528205969:pipeline/4697566D-Rental-Pipeline-V2/execution/0z9kck58arqm
```

11A. View the MLflow App run

The training step now sends its XGBoost parameters, validation metrics, and model artifact to the SageMaker MLflow App. The code below creates a presigned MLflow App URL so you can open the experiment in the MLflow UI.

In [18]:
```python
# Open the SageMaker MLflow App UI
# ============================================================
# AWS CLIENTS
# ============================================================

sm_client = boto3.client("sagemaker", region_name=REGION)
s3_client = boto3.client("s3", region_name=REGION)
s3_resource = boto3.resource("s3", region_name=REGION)
```

```python
mlflow_url_response = sm_client.create_presigned_mlflow_app_url(
    Arn=MLFLOW_APP_ARN
)

mlflow_ui_url = mlflow_url_response["AuthorizedUrl"]

print("MLflow App:", MLFLOW_APP_NAME)
print("MLflow Experiment:", MLFLOW_EXPERIMENT_NAME)
print("\nOpen this URL in your browser to view the MLflow run:")
print(mlflow_ui_url)
```

```
MLflow App: iti113-26s1-team08-mlflow-app
MLflow Experiment: ITI113/Team08/Experiment

Open this URL in your browser to view the MLflow run:
```


```
https://app-WLHEEZL7YYAH.mlflow.sagemaker.ap-southeast-1.app.aws/auth?authToken=eyJhbGciOiJIUzI1NiJ9.eyJh
dXRoVG9rZW5JZCI6IjVBWVlXUyIsImZhc0NyZWRlbnRpYWxzIjoiQWdWNHRMTlc5NHl2RkIzMWpJOW1qbGZHOTJFNzJkMm5ERWpZQjZVW
TNBNkpHWmtBWHdBQkFCVmhkM010WTNKNWNIUnZMWEIxWW14cFl5MXJaWGtBUkVFM1dUSlhkalZPUzJ4bWJqVndSVnB0SzBsdksxWmhTSF
ZYWVdGeVdWcEdOVTUwY0cwd1UydHdNMk5ZZVU1NGVEaDFXV2hJTVVReldHVmtOM1ZCYVhONlp6MDlBQUVBQjJGM2N5MXJiWE1BVUdGeWJ
qcGhkM002YTIxek9tRndMWE52ZFhSb1pXRnpkQzB4T2pNNU5qa3hNemN6TnpJMU5EcHJaWGt2WVRBNU1XRmhNRE10TnprMU5TMDBaakF5
TFdJMVpHWXRaVE5oTlRNd1pXSmlaVGcxQUxnQkFnRUFlT0thVkkrUUdqak5TNEo0TUhCNk91SlA3UGFLdlRHSG9tY2kveDlrZTJiekFhO
XM1b2h2UkRUK2RoekU4Ym03REhJQUFBQitNSHdHQ1NxR1NJYjNEUUVIQnFCdk1HMENBUUF3YUFZSktvWklodmNOQVFjQk1CNEdDV0NHU0
FGbEF3UUJMakFSQkF5Vi9qUS9aU3ZzMS9KZDIyWUNBUkNBTzZaUkJITUo4M1JPOGpvNXkzak5OSTlzaTVwNUc0QUdQSWtQRmYvWG5IZEE
2RDVnUG1RaE9tRklDTDNNWUVZeHM1aHhrYlg5cHo0TW9kdlpBZ0FBRUFERmtXSXVWWHNoSCsrMlBlQjIrNnNqcVBYT2hZYjA1dW5aaTJJ
dGUzWjVkQjQvS1ZxY0pPOUFBQWpBRFJzK2JXVC8vLy8vQUFBQUFRQUFBQUFBQUFBQUFBQUFBQUFBUUFBQTlNRlBrSld1dnZtanRCVzM3MVJGZ
jJsbE5JSmFLZlh6RXg3OG40Rzh2NHA5cW9ycWwyemZXWXYzYlovczV5N1lpWE4zenVzajUzNndodXVWRW9TNWVIZExRNkE0R05NbEErVD
Q2elIyaUsyT1d1WkhpbWZFT2hYbmxQcnpHSld2NENHUHVrM1BEMXNVTUo3T2xhWVlwSWpBRUxRTEF6ZURRR1VEWTdyWDBFSHFaOFUvYUZ
1QlNWbUx4eEtKRCthTUtWQXJHektONWRKbFJpd2YwcnRQSWFZdXpLTSthS0tDVlZmZDVmUjNZS0FMZ0FrYnM3aW1CbnhKbE84SFdReDhv
d2Y3cnNnaXRDL2pHbFdqUmNIQ3dtZmZNbzhjM1NlT3liWjFaNDBUZlJRS09SSDIvMldmYlpxelkweFk0dXVrMG16aVVjRlRSSCtVaUNTb
0RacWFPSmhybjljeWVuNWxrZ3lOOUdLbFd4QXQvc3Z6eko4YzZOVWgxZ0FhQ2gvS09tUTNzZWsxZWNsaE1ubnRkSDRQZWJPN2xmV285cD
YrL2V2T2lRVllqYk5qaStNMXR4VCtxdHZtcjYxbHR1bldwSWJ4SnJHOFUyeHRmSUVmK08zNnE3eVJkTzZPbHdaRjZ0L09YSjdEeDUreXF
RNFlVRVMwcXIweHhhbWhNMjZjZTcyVVY5ZVVJdWZXdzdHT0srendMOXBZTVY2SHBaS25TRDdnbFRIaXp5N1cxK1QyRWlQYTBqbEhFSWt2
UzZXU3dkcTB6UC9mK3U2WkFYMUlDMDY2UmRxaGdtTnRhczVUelduZ3VRRXFCekJEaFVZS1BmU3NiNkNKZ01MMnNjeGIrOExMYkpoQkVzL
0QvUmFFZ25INnlEOExicFprUWUrZTFXVmhpaTNsOEVCOGh0T2QrMGFkS2JJU2NKUWxLYmNVSFZlRnB1a0VPQ0x6N1FzY3EwS3ZZSGs3Y2
d1ZEF3ODBrUjhtRUd5YjZlbFlUQmtreWpnK3dEWVoyU241U09tSHZybUZTcjdiS0pCbnFmNWdxdzlBMlBJUDA0OUxPVWdzekJVb2hhSDd
lTXZGcmlTcW1PNzdiZW12MGhud3hDbkdiTEMzVDZ3SVZzREVFWjBXanpTQ3B6N3YzZG8yWG4vTVVXbVlScUZNdjNTYnUwYTZ1Q0cxRFM4
Q1V0R0wzbXJrZktOQW9rV3lvMW9uejk2SXBQU2tUS2NqMGxHaGExK0tNWisyVTVhWW90b3hsSUxQREMyTDBzYm1KWUErMmNVcG55T0RPU
XkxSW11Ym91dEM0MkxoNzQrNjQwZlQ4Q1dSWW4rTXhSblAyUjAwT3RIdlM1VUw5VlNhaWNPbXhMVVNDRGtOS3ZmanJmWXpjdGVvN01jdD
MrbG5UQ29nT0QwR20xZ242MXlpUTFFMWhldWNaeVJRdWJxZGhUVlR6STd1SGJwVm8wV2VRU0NBb2hwTSt5V2FQbmRMYkt3d1VEU1JKRkR
jY3JxMU5YQjdVa21VejFMVVgvSHBQc0dTVFlhcUZZRytvREVxUGNDQkViZmNQbDJkYktCalpKUndBUWhNcmpaOTdIWUFhTmE2eWlZSS9X
cU5IMUl0OXhxdlZ3ZnFQN0ZEWEVLOWdiRDl2cjk4M1hSRnVGU0x0U2lMSldaTUpMVUg2UEpxNXhDSXc5SStvczV0ZFZvNjRhWUxGV3hhV
2F2VjNZdy9wSU1NZ0x1alpMc1R1RzB6U0VBdkIweHd5YmEvSVQ3TElFcThVQTUzQUJuTUdVQ01RQ0hta1dMNlRBSTVJM1o3YjNpck1Yen
VNK3owMWhhOEFaa0RQUjk2SUJlOXZjRjBSU2RZV3ArNytnS1ZBVTNiblVDTUdkdVc2QUROeEJSWmhQempIZ3QzLzdYd1dFL2lpY0twV3F
tVWViUVZvem9mSWtaMHc3WjZLWXVlM3NJallkZEx3PT0iLCJjaXBoZXJUZXh0IjoiQVFJQkFIamltbFNQa0JvNHpVdUNlREJ3ZWpyaVQr
ejJpcjB4aDZKbkl2OGZaSHRtOHdHN0FaWjBqVVV6ZGZpSjgrRVZxYXRUQUFBQW9qQ0Jud1lKS29aSWh2Y05BUWNHb01HUk1JR09BZ0VBT
UlHSUJna3Foa2lHOXcwQkJ3RXdIZ1lKWUlaSUFXVURCQUV1TUJFRURKaDRWYVFUS0k5dnpJeXJjd0lCRUlCYkVzbjZ1dHJVNzRiam9zSH
M3M2FGbTVrSlMwYTJucHV5a0gweGdScGxXeU9ZYkNnUjBtNEZ1UzdoKzI5UGdnMjhDTWtQNXZkbFJoa0pPR1FVaVZ6WkVJYXljNTk3OGI
vb3BVR2pUWWxJNmd2UGNrUlYvZ2p4Vm5YbTFBPT0iLCJzdWIiOiJhcm46YXdzOnNhZ2VtYWtlcjphcC1zb3V0aGVhc3QtMTowNDQ1Mjgy
MDU5Njk6bWxmbG93LWFwcC9hcHAtV0xIRUVaTDdZWUFIIiwiaWF0IjoxNzg3NTcxMTczLCJleHAiOjE3ODc1NzE0NzN9.piiS3eHQbji3
ayqhMS--d03VahAV_HiXVcaaU46aqu0
```


13. Deploy the latest approved model

Because the registered model now contains the custom inference specification, deployment through the Model Registry can use that specification.

In [19]:
```python
import boto3

sm = boto3.client("sagemaker", region_name=region)

response = sm.list_model_packages(
    ModelPackageGroupName=MODEL_PACKAGE_GROUP,
    SortBy="CreationTime",
    SortOrder="Descending",
    MaxResults=5
)

for p in response["ModelPackageSummaryList"]:
    print(
        p["ModelPackageArn"],
        p["ModelPackageStatus"],
        p.get("ModelApprovalStatus"),
```

```
        p["CreationTime"]
    )
```

```
arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/23 Completed Approved
2026-08-21 14:42:19.108000+00:00
arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/22 Completed Approved
2026-08-21 14:20:35.688000+00:00
arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/21 Completed Approved
2026-08-21 09:33:23.071000+00:00
arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/20 Completed Approved
2026-08-20 10:48:56.069000+00:00
arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/19 Completed Approved
2026-08-20 09:50:02.030000+00:00
```

In [20]:

```
from sagemaker import ModelPackage

deploy_session = sagemaker.Session()
sm_client = deploy_session.sagemaker_client

response = sm_client.list_model_packages(
    ModelPackageGroupName=MODEL_PACKAGE_GROUP,
    ModelApprovalStatus="Approved",
    SortBy="CreationTime",
    SortOrder="Descending",
    MaxResults=1,
)
packages = response["ModelPackageSummaryList"]
if not packages:
    raise RuntimeError(f"No Approved Model Package found in {MODEL_PACKAGE_GROUP}")

latest = packages[0]
model_package_arn = latest["ModelPackageArn"]
model_package_version = latest["ModelPackageVersion"]
endpoint_name = f"sgp-rental-price-endpoint-v2-{model_package_version}"

print("Model package:", model_package_arn)
print("Endpoint:", endpoint_name)

model_from_registry = ModelPackage(
    role=role,
    model_package_arn=model_package_arn,
    sagemaker_session=deploy_session,
)

predictor = model_from_registry.deploy(
    initial_instance_count=1,
    instance_type=INFERENCE_INSTANCE,
    endpoint_name=endpoint_name,
)
print("Deployment successful:", endpoint_name)
```

```
/opt/conda/lib/python3.12/site-packages/sagemaker/model.py:347: SageMakerV2DeprecationWarning: ModelPacka
ge is part of the SageMaker Python SDK v2, which is on path of deprecation. In v3, use `ModelBuilder` (`f
rom sagemaker.serve import ModelBuilder`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
  warn_v2_deprecation(
WARNING:sagemaker.deprecations:ModelPackage is part of the SageMaker Python SDK v2, which is on path of d
eprecation. In v3, use `ModelBuilder` (`from sagemaker.serve import ModelBuilder`).
See https://github.com/aws/sagemaker-python-sdk/blob/master/migration.md for the migration guide. Set SAG
EMAKER_SUPPRESS_V2_WARNING=1 to silence this warning.
INFO:sagemaker:Creating model with name: RentalPriceModelGroupV2-2026-08-24-11-34-10-796
Model package: arn:aws:sagemaker:ap-southeast-1:044528205969:model-package/RentalPriceModelGroupV2/23
Endpoint: sgp-rental-price-endpoint-v2-23
INFO:sagemaker:Creating endpoint-config with name sgp-rental-price-endpoint-v2-23
INFO:sagemaker:Creating endpoint with name sgp-rental-price-endpoint-v2-23
-
-
-
-
-
-
!
Deployment successful: sgp-rental-price-endpoint-v2-23
```

In [21]:
```
#check status of pipeline

import boto3

sm_client = boto3.client("sagemaker", region_name=region)  # same `region` var your notebook already use

response = sm_client.list_pipelines(PipelineNamePrefix="4697566D")
for p in response["PipelineSummaries"]:
    print(p["PipelineName"], "|", p["PipelineArn"], "|", p.get("LastExecutionStatus", "N/A"))
```

```
4697566D-Rental-Pipeline-V2 | arn:aws:sagemaker:ap-southeast-1:044528205969:pipeline/4697566D-Rental-Pipe
line-V2 | N/A
4697566D-Rental-Pipeline-V1 | arn:aws:sagemaker:ap-southeast-1:044528205969:pipeline/4697566D-Rental-Pipe
line-V1 | N/A
```

Status of Containers

In [22]:
```
MODEL_PACKAGE_ARN=model_package_arn
package = sm.describe_model_package(
    ModelPackageName=MODEL_PACKAGE_ARN
)

print("Status:")
print(package["ModelPackageStatus"])

print("\nApproval:")
print(package.get("ModelApprovalStatus"))

print("\nContent types:")
print(package.get("InferenceSpecification", {}).get("SupportedContentTypes"))

print("\nResponse types:")
print(package.get("InferenceSpecification", {}).get("SupportedResponseMIMETypes"))

print("\nContainers:")
for container in package.get("InferenceSpecification", {}).get("Containers", []):
    print("\nImage:")
    print(container.get("Image"))

    print("ModelDataUrl:")
    print(container.get("ModelDataUrl"))

    print("Environment:")
    print(container.get("Environment"))
```

```
Status:
Completed

Approval:
Approved

Content types:
['application/json', 'text/csv']

Response types:
['application/json', 'text/csv']

Containers:

Image:
121021644041.dkr.ecr.ap-southeast-1.amazonaws.com/sagemaker-xgboost:1.7-1
ModelDataUrl:
s3://sagemaker-ap-southeast-1-044528205969/pipelines-3pqkkqxdtoby-xgboost-RepackModel-kB3UrvYxW6/output/m
odel.tar.gz
Environment:
{'SAGEMAKER_CONTAINER_LOG_LEVEL': '20', 'SAGEMAKER_PROGRAM': 'inference.py', 'SAGEMAKER_REGION': 'ap-sout
heast-1', 'SAGEMAKER_SUBMIT_DIRECTORY': 's3://sagemaker-ap-southeast-1-044528205969/sagemaker-xgboost-202
6-08-21-14-25-54-538/sourcedir.tar.gz'}
```

** Post Deployment Check, JSON inference and Test Data Input of endpoint **

In [23]:
```
import sys
import os

sys.path.insert(0, "src")
```

```python
print("Python:", sys.version)

import inference

print("inference.py imported successfully")
print("Feature count:", len(inference.FEATURE_NAMES))
print("Features:", inference.FEATURE_NAMES)
```

```
Python: 3.12.13 | packaged by conda-forge | (main, Mar  5 2026, 16:50:00) [GCC 14.3.0]
inference.py imported successfully
Feature count: 31
Features: ['bedroom', 'area_sm', 'lease_year', 'lease_month', 'dist_1', 'dist_2', 'dist_3', 'dist_4', 'di
st_5', 'dist_6', 'dist_7', 'dist_8', 'dist_9', 'dist_10', 'dist_11', 'dist_12', 'dist_13', 'dist_14', 'di
st_15', 'dist_16', 'dist_17', 'dist_18', 'dist_19', 'dist_20', 'dist_21', 'dist_22', 'dist_23', 'dist_2
5', 'dist_26', 'dist_27', 'dist_28']
```

In [24]:

```python
import json
import boto3
import pandas as pd
import numpy as np

ENDPOINT_NAME = endpoint_name  # update to match your redeployed endpoint name
REGION = region  # use the dynamically detected region from setup, not a hardcoded literal

DISTRICT_CATEGORIES = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,
                       17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28]

#Payload only sends the bedroom,area_sm, lease_month, lease_year and key which only sends the selected d

def build_payload(bedroom: int, area_sm: int, lease_month: int, lease_year: int, district: int) -> dict:

    if district not in DISTRICT_CATEGORIES:
        raise ValueError(f"District {district} is not a recognized category.")
    if float(bedroom) != int(bedroom):
        raise ValueError(f"bedroom must be a whole number, got {bedroom}")
    if float(area_sm) != int(area_sm):
        raise ValueError(f"area_sm must be a whole number, got {area_sm}")
    return {
        "bedroom": int(bedroom),
        "area_sm": int(area_sm),
        "lease_month": lease_month,
        "lease_year": lease_year,
        f"dist_{district}": 1,
    }

# Return from predictor is in full integer sum e.g. $5,050 instead of 5.05 thousand

def predict(runtime, payload: dict) -> str:

    response = runtime.invoke_endpoint(
        EndpointName=ENDPOINT_NAME,
        ContentType="application/json",
        Accept="application/json",
        Body=json.dumps(payload),
    )
    result = json.loads(response["Body"].read().decode("utf-8"))
    return result[0]

#text / csv is sent instead of application/json

def predict_csv(runtime, payload: dict) -> str:

    full_payload = {
        "bedroom": payload["bedroom"],
        "area_sm": payload["area_sm"],
        "lease_year": payload["lease_year"],
        "lease_month": payload["lease_month"],
    }
    for d in DISTRICT_CATEGORIES:
        key = f"dist_{d}"
        full_payload[key] = payload.get(key, 0)

    csv_body = ",".join(str(v) for v in full_payload.values())  # values only, no header
```

```python
    response = runtime.invoke_endpoint(
        EndpointName=ENDPOINT_NAME,
        ContentType="text/csv",
        Accept="application/json",
        Body=csv_body,
    )
    result = json.loads(response["Body"].read().decode("utf-8"))
    return result[0]

#Formatted output back into plain float

def parse_dollar_string(s: str) -> float:

    return float(s.replace("$", "").replace(",", ""))


def run_manual_test_cases():
    runtime = boto3.client("sagemaker-runtime", region_name=REGION)
    """
    test_cases = [
        {"bedroom": 3, "area_sm": 90, "lease_month": 6, "lease_year": 2024, "district": 5},
        {"bedroom": 5, "area_sm": 200, "lease_month": 1, "lease_year": 2024, "district": 10},
        {"bedroom": 1, "area_sm": 40, "lease_month": 12, "lease_year": 2023, "district": 15},
    ]
    """
    test_cases = [
        {"bedroom": 3, "area_sm": 90, "lease_year": 2024, "lease_month": 6, "district": 5},
        {"bedroom": 5, "area_sm": 200, "lease_year": 2024, "lease_month": 1, "district": 10},
        {"bedroom": 1, "area_sm": 40, "lease_year": 2023, "lease_month": 12, "district": 15},
    ]


    for i, case in enumerate(test_cases, 1):
        payload = build_payload(**case)
        predicted_rent = predict(runtime, payload)  # already formatted
        print(f"Test case {i}: {case}")
        print(f"  Predicted monthly rent: {predicted_rent}\n")

#Sample real rows and compare model prediction vs actual recorded rent.

def run_real_sample_test(csv_path: str = "rental_2125.csv", n: int = 5, seed: int = 42):

    df = pd.read_csv(csv_path)
    df = df.dropna(subset=["bedroom", "area_sm", "lease date", "district", "mth rent"])
    df = df[df["district"].isin(DISTRICT_CATEGORIES)]
    df["lease_year"] = pd.to_datetime(df["lease date"]).dt.year
    df["lease_month"] = pd.to_datetime(df["lease date"]).dt.month

    sample = df.sample(n=min(n, len(df)), random_state=seed)
    runtime = boto3.client("sagemaker-runtime", region_name=REGION)

    print(f"\n{'='*60}\nComparing predictions against real recorded rents\n{'='*60}")
    for _, row in sample.iterrows():
        payload = build_payload(
            bedroom=int(row["bedroom"]),
            area_sm=int(round(row["area_sm"])),
            lease_year=int(row["lease_year"]),
            lease_month=int(row["lease_month"]),
            district=int(row["district"]),
        )
        predicted_rent_str = predict(runtime, payload)  # already formatted
        predicted_monthly_rent = round(parse_dollar_string(predicted_rent_str))
        actual_monthly_rent = round(float(row["mth rent"]) * 1000)




        print(f"District {int(row['district'])}, {int(row['bedroom'])}BR, {row['area_sm']}sqm")
        print(f"  Predicted:{predicted_rent_str}  |  Actual: ${actual_monthly_rent:,.0f}  "
              f"|  Diff: ${predicted_monthly_rent - actual_monthly_rent:,.0f}")


if __name__ == "__main__":
    print("Running manual test cases (application/json)...\n")
    run_manual_test_cases()
```

```
    print("\nSame prediction via text/csv, for comparison...")
    runtime = boto3.client("sagemaker-runtime", region_name=REGION)
    payload = build_payload(bedroom=3, area_sm=90, lease_year=2024, lease_month=6, district=5)
    csv_result = predict_csv(runtime, payload)
    print(f"  Predicted monthly rent (via CSV): {csv_result}")
```

```
Running manual test cases (application/json)...

Test case 1: {'bedroom': 3, 'area_sm': 90, 'lease_year': 2024, 'lease_month': 6, 'district': 5}
  Predicted monthly rent: 4995

Test case 2: {'bedroom': 5, 'area_sm': 200, 'lease_year': 2024, 'lease_month': 1, 'district': 10}
  Predicted monthly rent: 10659

Test case 3: {'bedroom': 1, 'area_sm': 40, 'lease_year': 2023, 'lease_month': 12, 'district': 15}
  Predicted monthly rent: 2664


Same prediction via text/csv, for comparison...
  Predicted monthly rent (via CSV): 4995
```

Testing with CSV input

In [25]:

```
#  Verify the endpoint with a sample prediction

import json
import boto3
import pandas as pd

ENDPOINT_NAME = endpoint_name  # update to match your redeployed endpoint name
REGION = "ap-southeast-1"

DISTRICT_CATEGORIES = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23

def build_sample_row(bedroom, area_sm, lease_year, lease_month, district):
    district_onehot = [1 if d == district else 0 for d in DISTRICT_CATEGORIES]
    row = [bedroom, area_sm, lease_year, lease_month] + district_onehot
    return ",".join(str(v) for v in row)

runtime = boto3.client("sagemaker-runtime")

sample_row = build_sample_row(bedroom=3, area_sm=90, lease_year=2024, lease_month=6, district=5)
print("Sample input row:", sample_row)

response = runtime.invoke_endpoint(
    EndpointName=endpoint_name,
    ContentType="text/csv",
    Body=sample_row,
)
prediction = response["Body"].read().decode("utf-8").strip()
print(f"Predicted monthly rent (in thousands, matching training scale): {prediction}")

# Sanity check: try a second, very different sample to confirm predictions vary sensibly
sample_row_2 = build_sample_row(bedroom=5, area_sm=200, lease_year=2024, lease_month=1, district=10)
response_2 = runtime.invoke_endpoint(EndpointName=endpoint_name, ContentType="text/csv", Body=sample_row
prediction_2 = response_2["Body"].read().decode("utf-8").strip()
print(f"Second sample prediction: {prediction_2}")
```

```
Sample input row: 3,90,2024,6,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
Predicted monthly rent (in thousands, matching training scale): [4995]
Second sample prediction: [10659]
```

In [26]:

```
import json
import boto3

runtime = boto3.client("sagemaker-runtime", region_name=region)

DISTRICTS = [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,25,26,27,28]

def build_csv_row(bedroom, area_sm, lease_year, lease_month, district):
    one_hot = [1 if d == district else 0 for d in DISTRICTS]
    return ",".join(map(str, [bedroom, area_sm, lease_year, lease_month] + one_hot))

def invoke_csv():
    body = build_csv_row(3, 90, 2024 ,6, 5)
```

```python
    response = runtime.invoke_endpoint(EndpointName=endpoint_name, ContentType="text/csv", Accept="appli
    return response["Body"].read().decode()

def invoke_json():
    payload = {"bedroom": 3, "area_sm": 90, "lease_year": 2024, "lease_month": 6, "district": 5}
    response = runtime.invoke_endpoint(EndpointName=endpoint_name, ContentType="application/json", Accep
    return response["Body"].read().decode()

print("CSV result :", invoke_csv())
print("JSON result:", invoke_json())
```

```
CSV result : [4995]
JSON result: [4995]
```

Modelsuccess criteria

- Processing completes.
- Training completes and produces `xgboost-model.ubj`.
- Evaluation produces `evaluation.json`.
- R^2 threshold condition passes.
- Model Registry contains a V2 model package.
- The model package includes the custom `inference.py` serving specification.
- Endpoint accepts `text/csv`.
- Endpoint accepts `application/json`.
- Compact JSON with `district` is converted to the one-hot feature vector expected by the model.

Pause for Endpoint Testing before removal

Testing the endpoint from browser

In [27]:
```
!pip install streamlit
```

```
Collecting streamlit
  Using cached streamlit-1.62.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /opt/conda/lib/python3.12/site-package
s (from streamlit) (6.2.2)
Requirement already satisfied: blinker<2,>=1.5.0 in /opt/conda/lib/python3.12/site-packages (from streaml
it) (1.9.0)
Requirement already satisfied: click<9,>=7.0 in /opt/conda/lib/python3.12/site-packages (from streamlit)
(8.2.1)
Requirement already satisfied: numpy<3,>=1.23 in /opt/conda/lib/python3.12/site-packages (from streamlit)
(1.26.4)
Requirement already satisfied: packaging>=20 in /opt/conda/lib/python3.12/site-packages (from streamlit)
(24.2)
Requirement already satisfied: pandas<4,>=1.4.0 in /opt/conda/lib/python3.12/site-packages (from streamli
t) (2.3.3)
Requirement already satisfied: pillow<13,>=7.1.0 in /opt/conda/lib/python3.12/site-packages (from streaml
it) (11.3.0)
```

```
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Using cached pydeck-0.9.3-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=5.26.1 in /opt/conda/lib/python3.12/site-packages (from strea
mlit) (6.31.1)
Requirement already satisfied: pyarrow!=25.0.0,<26,>=7.0 in /opt/conda/lib/python3.12/site-packages (from
streamlit) (21.0.0)
Requirement already satisfied: requests<3,>=2.27 in /opt/conda/lib/python3.12/site-packages (from streaml
it) (2.34.2)
Collecting toml<2,>=0.10.1 (from streamlit)
  Using cached toml-0.10.2-py2.py3-none-any.whl.metadata (7.1 kB)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /opt/conda/lib/python3.12/site-packages (f
rom streamlit) (4.16.0)
Requirement already satisfied: starlette<2,>=0.46.0 in /opt/conda/lib/python3.12/site-packages (from stre
amlit) (0.52.1)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /opt/conda/lib/python3.12/site-packages (from stream
lit) (0.51.0)
Requirement already satisfied: httptools<1,>=0.6.3 in /opt/conda/lib/python3.12/site-packages (from strea
mlit) (0.8.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /opt/conda/lib/python3.12/site-packages (from streamli
t) (4.14.2)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /opt/conda/lib/python3.12/site-packages (fr
om streamlit) (0.0.32)
Requirement already satisfied: websockets<17,>=12.0.0 in /opt/conda/lib/python3.12/site-packages (from st
reamlit) (16.1.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /opt/conda/lib/python3.12/site-packages (from st
reamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /opt/conda/lib/python3.12/site-packages (from stream
lit) (6.0.0)
Requirement already satisfied: jinja2 in /opt/conda/lib/python3.12/site-packages (from altair!=5.4.0,!=5.
4.1,<7,>=5.0.0->streamlit) (3.1.6)
Requirement already satisfied: jsonschema>=3.0 in /opt/conda/lib/python3.12/site-packages (from altair!=
5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=2.4.0 in /opt/conda/lib/python3.12/site-packages (from altair!=
5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (2.24.0)
Requirement already satisfied: idna>=2.8 in /opt/conda/lib/python3.12/site-packages (from anyio<5,>=4.0.0
->streamlit) (3.18)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.12/site-packages (from pa
ndas<4,>=1.4.0->streamlit) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.12/site-packages (from pandas<4,>=
1.4.0->streamlit) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in /opt/conda/lib/python3.12/site-packages (from pandas<4,>
=1.4.0->streamlit) (2026.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /opt/conda/lib/python3.12/site-packages (from
requests<3,>=2.27->streamlit) (3.4.9)
Requirement already satisfied: urllib3<3,>=1.26 in /opt/conda/lib/python3.12/site-packages (from requests
<3,>=2.27->streamlit) (1.26.20)
Requirement already satisfied: certifi>=2023.5.7 in /opt/conda/lib/python3.12/site-packages (from request
s<3,>=2.27->streamlit) (2026.6.17)
Requirement already satisfied: h11>=0.8 in /opt/conda/lib/python3.12/site-packages (from uvicorn<1,>=0.3
0.0->streamlit) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.12/site-packages (from jinja2->a
ltair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (3.0.3)
Requirement already satisfied: attrs>=22.2.0 in /opt/conda/lib/python3.12/site-packages (from jsonschema>
=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (25.4.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /opt/conda/lib/python3.12/site-pac
kages (from jsonschema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /opt/conda/lib/python3.12/site-packages (from jsons
chema>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (0.37.0)
Requirement already satisfied: rpds-py>=0.7.1 in /opt/conda/lib/python3.12/site-packages (from jsonschema
>=3.0->altair!=5.4.0,!=5.4.1,<7,>=5.0.0->streamlit) (0.27.1)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.12/site-packages (from python-dateutil>
=2.8.2->pandas<4,>=1.4.0->streamlit) (1.17.0)
Using cached streamlit-1.62.0-py3-none-any.whl (10.5 MB)
Using cached pydeck-0.9.3-py2.py3-none-any.whl (11.4 MB)
Using cached toml-0.10.2-py2.py3-none-any.whl (16 kB)
Installing collected packages: toml, pydeck, streamlit
   ━━━━━━━━━━━━━╺━━━━━━━━━━━━━━━━━━━━━━━━━━ 1/3 [pydeck]
   ━━━━━━━━━━━━━╺━━━━━━━━━━━━━━━━━━━━━━━━━━ 1/3 [pydeck]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
   ━━━━━━━━━━━━━━━━━━━━━━━━━━╺━━━━━━━━━━━━━ 2/3 [streamlit]
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3/3 [streamlit]

Successfully installed pydeck-0.9.3 streamlit-1.62.0 toml-0.10.2
```

In [28]:

```
!pip install pyngrok
```

```
Collecting pyngrok
  Using cached pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: PyYAML>=5.1 in /opt/conda/lib/python3.12/site-packages (from pyngrok) (6.
0.3)
Using cached pyngrok-8.1.2-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.2
```

In [29]:

```
from pyngrok import ngrok

ngrok.set_auth_token("3I4yHHTo3NjAEeAFjTDHKqhSKfR_2HJX8mLz24XirqzEqo6q")
```

```
INFO:pyngrok.process:Updating authtoken for default "config_path" of "ngrok_path": /home/sagemaker-user/.
config/ngrok/ngrok
```

In [30]:

```
import time

print("Application will pause for 3 mins before tearing down endpoint")
time.sleep(120)
print("Program resumed to clean up endpoint")
```

```
Application will pause for 3 mins before tearing down endpoint
Program resumed to clean up endpoint
```

Clean Up*

In [31]:

```
# Clean up: delete the endpoint  then VERIFY deletion

endpoint_desc = sm_client.describe_endpoint(EndpointName=endpoint_name)
endpoint_config_name = endpoint_desc["EndpointConfigName"]

sm_client.delete_endpoint(EndpointName=endpoint_name)
print(f"Delete requested for endpoint: {endpoint_name}")

sm_client.delete_endpoint_config(EndpointConfigName=endpoint_config_name)
print(f"Delete requested for endpoint config: {endpoint_config_name}")

# Verify the endpoint is actually gone — describe_endpoint should now raise a ClientError
from botocore.exceptions import ClientError

try:
    sm_client.describe_endpoint(EndpointName=endpoint_name)
    print("WARNING: endpoint still exists — deletion may not have completed yet. Wait a few seconds and
except ClientError as e:
    if "Could not find endpoint" in str(e) or "ValidationException" in str(e):
        print(f"Verified: endpoint '{endpoint_name}' no longer exists.")
    else:
        raise

try:
    sm_client.describe_endpoint_config(EndpointConfigName=endpoint_config_name)
    print("WARNING: endpoint config still exists.")
except ClientError as e:
    if "Could not find endpoint configuration" in str(e) or "ValidationException" in str(e):
        print(f"Verified: endpoint config '{endpoint_config_name}' no longer exists.")
    else:
        raise
```

```
Delete requested for endpoint: sgp-rental-price-endpoint-v2-23
Delete requested for endpoint config: sgp-rental-price-endpoint-v2-23
WARNING: endpoint still exists — deletion may not have completed yet. Wait a few seconds and re-check.
Verified: endpoint config 'sgp-rental-price-endpoint-v2-23' no longer exists.
```

In [ ]: